

Creativity in Olympic Bouldering: Exploring the Role of Climbing Level and Route Previewing

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In sports, creativity refers to the ability to make decisions that are original and unexpected, yet effective. Given the limited research on climbers' creativity and its underlying mechanisms, this study explored the role of climbing level and route previewing on climbers' creativity and their ability to find creative climbing solutions. Twenty-eight climbers (14 advanced and 14 elite) were tasked with climbing an Olympic boulder that offered two climbing options: an easily identifiable option beyond their climbing levels, and a creative option that, though less identifiable, was feasible for both groups. Results revealed that elite climbers were more successful in completing the boulder, attained higher creativity scores, and more often used the creative option. Despite shorter preview durations, they had fewer holds without fixations and more fixations at key holds of the creative option. Overall, findings indicate that creativity in climbing is associated with skill level and behavior during route previewing.

Keywords: cognition, decision making, eye-tracking, gaze behavior, rock climbing

For many decades, the investigation of creativity has constituted a seminal research topic within the field of psychology (Runco, 2007; Sternberg & Lubart, 1999). A contemporary and widely acknowledged multidimensional, integrative definition of creativity originates from Sternberg and Lubart (1999, p. 3), who defined creativity as an individual's ability "to produce work that is both novel (original, unexpected) and appropriate (useful)." This foundational understanding of creativity is rooted in the theoretical framework of divergent thinking (Guilford, 1967), which involves developing multiple ideas and solutions to a given situation, rather than converging on a single response (Sternberg & Lubart, 1999). Divergent thinking is conceptualized as a cognitive process that fosters flexibility and openness, encouraging the exploration of diverse perspectives. This cognitive process is particularly relevant to creative decision making and the generation of original and unexpected solutions (Runco & Acar, 2019).

Within the context of sport, interest in the relationship between creativity and performance has substantially increased over the past two decades (e.g., de Jooode et al., 2023; Fardilha & Allen, 2020; Memmert, 2015). Creativity in sport is conceptualized as an athlete's ability to make decisions and generate motor solutions that are unique, original, and unexpected, while still maintaining effective (Memmert, 2011, 2013; Memmert & Roth, 2007). To date, a considerable body of research has emphasized the importance of creativity in achieving optimal performance, particularly in

open, fast-paced team sports with high time constraints (Kempe & Memmert, 2018; Memmert & Roca, 2019; Roca et al., 2018). In recent years, research has also begun to explore the role of creativity in individual, less-open sports that do not involve direct opponents. Olympic bouldering is one such sport in which creativity plays a key role in achieving success (Künzell et al., 2021).

Olympic bouldering is a climbing discipline in which athletes must solve a series of short climbing routes on artificial walls, known as boulders (Augste et al., 2021; Medernach & Memmert, 2021). These boulders present climbers with the unique challenge of mastering physically strenuous, complex, and creative climbing movements within a time limit of 4–5 min (Medernach, Sanchez, et al., 2024). Climbers can make multiple attempts within the limited time given for each boulder, allowing them to adapt and refine their strategies and motor solutions based on prior attempts. However, their ranking in each competition is determined by the number of successfully completed boulders and the number of attempts required to attain these completions (see Hatch & Leonardon, 2024). Consequently, success in Olympic bouldering relies on climbers' ability to make effective decisions and quickly develop efficient climbing solutions (Medernach et al., 2025a).

Strategic planning of attempts prior to climbing is considered a critical factor in achieving effective decision making and developing optimal climbing solutions (Medernach, Henz, et al., 2024; Sanchez et al., 2019). Particularly during the route previewing phase, climbers develop potential climbing strategies by cognitively processing climbing movements and gathering functional information from visual cues of climbing holds (Medernach, Sanchez, et al., 2024; Sanchez et al., 2024). Climbers' behavior during route previewing is positively influenced by their domain-specific expertise (Medernach et al., 2023), which contributes to an extensive cognitive-motor knowledge (Mangan et al., 2024). In fact, the ability of climbers to quickly identify effective climbing

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solutions depends on their repertoire of climbing movements, allowing them to quickly interpret visual sensory input, recognize meaningful movement patterns, and anticipate climbing movements (Henz et al., 2025b; Medernach et al., 2025b).

In the endeavor to gain a deeper understanding of climbers' decision making and strategic planning skills, research has only recently begun to explore the role of creativity in climbing. Künzeli et al. (2021) were among the first to describe creativity as a key factor in achieving optimal climbing performance. In particular, the authors examined a series of IFSC (International Federation of Sport Climbing) world cups and observed that, following unsuccessful attempts, developing new, creative climbing solutions more often contributed to successful boulder completion than repeating strategies used in previous attempts. More recently, Henz et al. (2024) identified the ability to make creative decisions as an essential skill of elite climbers with routesetting expertise, further substantiating creativity as a key performance factor in Olympic bouldering. In the context of Olympic bouldering, creativity involves unconventional climbing solutions—such as 360° movements or the use of shortcuts to bypass movement sequences—that deviate from the apparent path typically intended by routesetters or expected by spectators (Medernach et al., 2025a).

Within the perceptual-cognitive framework of Olympic bouldering, creativity is conceptualized as a cognitive subprocess that precedes decision making and is shaped by perceptual information gathered during route previewing (Medernach & Memmert, 2021; Medernach, Sanchez, et al., 2024). Consistent with decision making processes in other sports (see Memmert & Roca, 2019), climbers initially anticipate and interpret a boulder based on prior experiences; subsequently, attention is directed toward relevant aspects of the boulder, after which various options—potentially including creative solutions—are evaluated, with one ultimately selected for motor execution. Extending Kahneman (2011) dual-process model of cognition to the context of Olympic bouldering, creative motor solutions primarily emerge from “slow thinking” processes, which are characterized by deliberate and analytical examination of climbing movements and hold configurations prior to climbing (Medernach et al., 2025a). In light of the conceptual distinction between “creative thinking” and “creative action” within the sporting context (Fardilha & Allen, 2020), creativity in Olympic bouldering is thus primarily associated with planned, compositional processes (see Marshall et al., 2024; Santos et al., 2016, 2020) of generating novel, yet effective solutions during route previewing.

Naturally, in certain scenarios, time constraints imposed by IFSC regulations compel climbers to rely more on intuitive, spontaneous, and improvised decisions during performance, bearing similarity to ad hoc adjustments made in responses to unpredictable and dynamic environments observed in team sports (Fardilha & Allen, 2020; Orth et al., 2017; Roca et al., 2018). However, recent research on Olympic bouldering emphasizes that climbers tend to eschew fast, intuitive decision making, instead striving to maintain analytical processing of task-relevant information and sensorimotor feedback derived from prior attempts (see Medernach et al., 2025a).

While empirical research on climbers' creativity remains sparse, considerable research has been conducted on creativity within the perceptual-cognitive framework, emphasizing the interplay of perceptual processes and cognitive mechanisms in the generation of novel, yet functionally appropriate motor solutions (Broadbent et al., 2015; Roca et al., 2021; Santos et al., 2016). To exemplify, expert soccer players have been found to rely on

postural cues and pattern recognition to effectively anticipate game situations (Roca et al., 2018, 2021), enabling them to make creative decisions that are unexpected and thus difficult for opponents to anticipate (Memmert & Roca, 2019). Research has also shown that athletes' focus of attention is a critical factor in creative decision making (e.g., Memmert, 2011). Specifically, a narrow attentional breadth constrains the amount of environmental information athletes perceive, thereby reducing their ability to identify creative motor solutions (Memmert & Furley, 2007; Roca et al., 2018). In contrast, a wide breadth of attention helps them to identify information that may be relevant for finding creative solutions (Furley et al., 2010; Memmert & Roca, 2019; Moraru et al., 2016). In this context, experts have also been found to be more successful at directing their attention toward more relevant aspects of tasks (Hüttermann et al., 2014), effectively filtering out irrelevant distractions (Furley et al., 2013), and less affected by inattention blindness when making decisions (Furley et al., 2010). The concept of inattention blindness, originally introduced by Rock et al. (1992), refers to the failure to perceive visible objects due to cognitive limitations in information processing and an inappropriate focus of attention.

To deepen our understanding of the factors underlying creativity in Olympic bouldering within the perceptual-cognitive framework, further investigation is required into two critical aspects that remain underexplored and warrant additional elucidation. First, while research in other sports suggests that creativity is—among other factors—closely linked to expertise (e.g., Fardilha & Allen, 2020; Memmert & Roca, 2019; Roca et al., 2021), it remains unclear whether higher climbing levels are inherently associated with enhanced creativity skills. In fact, given the variation in boulder styles within Olympic bouldering (see Augste et al., 2021; Henz et al., 2025a), some climbers may, for instance, demonstrate proficiency in athletic-style boulders—primarily reliant on upper body strength and power, while requiring minimal creative skill—and consequently be characterized by high climbing levels on the IRCRA scale (i.e., International Rock Climbing Research Association). Given the paucity of empirical research on climbers' creativity, there is currently no evidence that climbing level is associated with creativity.

Second, assuming a positive relationship between climbing level and creativity, and given the critical role of route previewing in developing efficient climbing solutions, it remains unclear whether route previewing serves as a contributing factor in enabling expert climbers to generate creative solutions. In light of research from other sports, highlighting the role of attentional focus and pattern recognition in the development of creative motor solutions, it remains unclear whether expert climbers similarly benefit from differences in attentional focus and more thorough processing of visual stimuli during route previewing to enhance their ability to identify creative climbing solutions.

Therefore, the present study explored the role of climbing level and route previewing as contributing factors to climbers' creativity and their ability to find creative climbing solutions. To this end, advanced and elite climbers were tasked with climbing an Olympic boulder that offered two climbing options: a visually apparent and easily identifiable option that exceeded their climbing levels, and a creative option that was less apparent, yet feasible for both groups and thus more effective. Extending the research by Künzeli et al. (2021), we hypothesized (Hypothesis 1: effect of climbing level) that elite climbers would demonstrate superior creativity skills and, consequently, be more likely to use the creative option to climb the boulder compared to advanced climbers. Building upon Hypothesis 1 and drawing on research in Olympic bouldering that

highlights the relationship between cognitive-behavioral processes during route previewing and climbing level (Medernach, Sanchez, et al., 2024), as well as evidence from other sports underpinning the role of visual search behaviors in generating creative motor solutions (Roca et al., 2018), we hypothesized (Hypothesis 2: effect of route previewing) that superior creativity skills among elite climbers would be associated with more efficient visual search strategies during route previewing.

Methods

Participants

To participate in the study, individuals were required to meet the following inclusion criteria: be at least 18 years old, be in good health with no recent injuries, and have a climbing level of at least 20 points (advanced level) on the IRCRA scale. To attain a largest possible sample size, climbers were approached in commercial gyms and invited to participate in the study. A total of 31 climbers who met the inclusion criteria agreed to participate. However, since only three were female, the study was conducted exclusively with male climbers to maintain sample consistency. This exclusion also served to reduce variability in physical and morphological characteristics among participants.

Ultimately, a total of 28 male climbers participated in the study. All participants provided written informed consent, and the study adhered to the ethical standards of the World Medical Association. Ethical approval was obtained from the University Ethics Committee (057/2020). To explore the role of climbing level as a factor contributing to climbers' creativity, participants were assigned to either the advanced (ADV; $n = 14$) or the elite (ELI; $n = 14$) group based on their IRCRA scores (see "Measures" section). This resulted in two ability groups with an equal number of participants (see Table 1). Specifically, the IRCRA scale is a reliable and valid tool widely used to classify climbing ability through a numerical system, thereby enabling the categorization of climbers into distinct ability groups (see Draper et al., 2016). In line with recent research on Olympic bouldering (e.g., Henz et al., 2025b; Medernach, Henz, et al., 2024), climbing level assessment included both participants' all-time highest climbing achievements (highest climbing level) and their climbing levels at the time of data collection (current climbing level).

Overview of the Study

Before the experiment began, participants were instructed to take a 48-hr rest, during which they should avoid nonessential physical

activities to prevent prefatigue from negatively affecting their climbing performance. Upon arriving at the climbing gym, they were interviewed to gather information about their climbing expertise. They received both verbal and written explanations about the purpose and procedures of the study. After having signed the consent form, their body weight and height were measured, and they engaged in their routine warm-up programs to ensure optimal physiological and psychological preparation.

Once they were warmed-up and confirmed to be ready to climb, participants were introduced to a novel boulder they had to climb in accordance with the IFSC rules for final rounds. This included a 2-min preview phase, followed by a 4-min climbing period (Hatch & Leonardon, 2024). The 2-min time limit for boulder previewing was monitored using a digital clock display. Participants were not allowed to view the boulder prior to the designated preview period or to physically rehearse it during the previewing phase. They were tested individually to avoid any exchange of information during the preview and to ensure they did not observe other participants during their attempts.

Prior to the boulder preview, a mobile eye-tracking device was mounted on the participants' head, followed by a calibration procedure to ensure the accuracy of the eye movement recording. Following completion of the calibration procedure, participants began to preview the boulder, during which their gaze behavior was assessed. Upon completing the preview, either by verbally informing the examiner or after the 2-min time limit had elapsed, participants were instructed to turn away from the wall to prevent any further visual exposure to the boulder.

Following the previewing phase, participants were tasked with climbing the boulder. The climbing procedure ended under one of three scenarios: when they successfully completed the boulder, when the 4-min time limit was reached, or when they decided not to make any further attempts. In a final step following the climbing procedure, participants took part in postclimbing interviews to gather relevant information about their behavior during the route preview and their climbing solutions.

Design of the Boulder

Two expert routesetters, with routesetting qualifications (European Qualifications Framework: Level 4), 10 years of climbing expertise, and elite climbing skills (≥ 27 IRCRA points), were tasked with setting the boulder. The boulder was newly designed and included climbing holds that participants were unfamiliar with (mix of brands with customized shapes). In accordance with the IFSC regulations, the boulder comprised a total of 12 holds; two starting footholds (Holds 1 and 2), two starting handholds (Holds 3 and 4),

Table 1 Personal Characteristics and Climbing Expertise of the ADV and ELI Groups

Variable (unit)	ADV group ($n = 14$)	ELI group ($n = 14$)	<i>t</i>	<i>p</i>	<i>r</i>
Age (years)	29.1 $\pm$ 5.6	28.1 $\pm$ 7.9	.361	.722	.073
Height (cm)	179.5 $\pm$ 5.5	178.9 $\pm$ 3.5	.367	.717	.065
Weight (kg)	68.9 $\pm$ 6.8	67.4 $\pm$ 5.8	.659	.516	.118
Highest climbing level (IRCRA score)	19.3 $\pm$ 0.9	25.5 $\pm$ 1.5	-13.197	<.001	.929
Current climbing level (IRCRA score)	18.7 $\pm$ 0.9	25.0 $\pm$ 1.1	-16.363	<.001	.953
Climbing experience (years)	3.0 $\pm$ 1.8	11.4 $\pm$ 5.2	-5.687	<.001	.734
Climbing competitions (number)	3.9 $\pm$ 4.8	14.9 $\pm$ 13.1	-2.948	.009	.487

Note. Results are presented as mean $\pm$ SD, followed by the *t*-test findings (*t*), the *p* values, and the effect sizes (*r*). IRCRA = International Rock Climbing Research Association; ADV = advanced; ELI = elite.

and eight additional holds (see Figure 1). The boulder was intentionally designed to offer two different climbing options. Participants were not provided with any information regarding the two options or their respective difficulty levels.

One option was easily identifiable and visually apparent; however, its difficulty, rated at 27 IRCRA points by the two routesetters, exceeded the climbers' skill levels. In this option, participants had to perform a dynamic movement from the starting position to grasp Hold 6, which was difficult to maintain. Hold 6 was marked with chalk to mislead them into believing that using this hold was essential for climbing the boulder. From Hold 6, they had to swing over to Hold 7, marking the second challenging movement of this option.

In contrast, the alternative option was shorter and less difficult, aligning with climbers' skill levels. Rated at 17 IRCRA points by the two routesetters, it was theoretically climbable for all participants, thus representing the more effective solution; however, this option was less apparent and more difficult to identify, making it a more original, unexpected, and thus creative solution. Specifically, participants had to bypass Holds 5 and 6 by balancing rightward from the starting position, placing their feet on Hold 7, and using the wall to stem upward and grasp Hold 8. However, Hold 8—a key hold in the creative option—was intentionally concealed on the right side of Hold 7, with both holds having the same color, thereby increasing the likelihood that Hold 8 would be overlooked during route previewing.

Measures

Personal Characteristics

Participants' ages were assessed based on the information provided in their consent forms (see Table 1). Body weight was measured in shorts and t-shirts to the nearest 0.5 kg using a mechanical scale (Soehnle Professional 6163). Body height was determined without shoes to the nearest 0.5 cm using a stadiometer (Seca 213).

Climbing Expertise

Participants' climbing expertise was assessed during the interviews conducted prior to the experiment. These assessments encompassed (a) their climbing levels on the IRCRA scale, (b) years of climbing experience, and (c) number of climbing competitions in which they had participated.

Climbing Performance and Time Management

Climbing performance and time management during climbing were assessed in situ by two experienced bouldering judges with climbing qualifications (European Qualifications Framework: Level 3). The judges were not informed about the group assignments of the participants. The variables measured for each participant included: (a) successful completion of the boulder (Top, coded as 0 = no; 1 = yes), (b) the highest hold reached during the best



Figure 1 — The boulder of the study. *Note.* The boulder offered two climbing options. In the easily identifiable option (27 International Rock Climbing Research Association [IRCRA] points), climbers had to make a dynamic movement from the starting position (Holds 1–4) to Hold 6 before swinging to Hold 7. In the creative option (17 IRCRA points), climbers shortcut the boulder by balancing from the starting position to the right, placing their feet on Hold 7 while using the wall to stem up and grasp Hold 8.

attempt, (c) the total number of climbing attempts made on the boulder, (d) the average climbing time per attempt, and (e) the average resting time between successive attempts. Climbing and resting times were averaged for each participant, and the mean score from both judges was retained as the final scores. The judges' independent ratings revealed perfect consistency for the Tops, highest reached climbing hold, and attempts ($\kappa=1.000$, $p<.001$), as well as high consistency for the average climbing and resting times ($\kappa\geq .924$, $p<.001$).

In accordance with the IFSC rules, a boulder was considered successfully completed when participants reached the designated finishing hold with both hands and in a controlled position. A climbing attempt was defined as the moment when all parts of the body left the ground. In each attempt, participants had to begin the boulder from the designated starting holds, requiring them to be in a controlled position with both hands and feet placed on the marked starting holds.

Effect of Climbing Level (Hypothesis 1)

Participants' creativity skills were assessed by three climbing experts using video recordings. Each of the experts had climbing qualifications (European Qualifications Framework: $\geq$ Level 4), over 10 years of climbing experience, and elite climbing skills (≥ 25 IRCRA points). The experts, like the judges, were not informed about the group assignments of the participants.

As an initial step, the three experts independently assigned a creativity score to each participant. Building upon prior research assessing creativity in team sports (e.g., Fardilha & Allen, 2020; Kempe & Memmert, 2018; Memmert & Roth, 2007), the participants' creativity in each attempt at the boulder was rated using a 5-point Likert scale (1 = *marginally creative*; 2 = *somewhat creative*; 3 = *moderately creative*; 4 = *very creative*; 5 = *highly creative*). Specifically, irrespective of whether participants chose the more creative option, the experts evaluated each attempt based on its overall originality and functional effectiveness (e.g., the use of a toe hook to maintain balance). The ratings for each attempt were averaged, and the mean scores from the three experts were used as the final creativity scores. The intraclass correlation coefficient revealed a high level of consistency among the three experts (movement characteristics: $r=.924$).

Furthermore, the experts also assessed whether participants used the more creative option (coded as 0 = no; 1 = yes) and determined the attempt in which participants altered their strategy to use this option. Experts' independent ratings revealed perfect consistency for both variables ($r=1.00$, $p<.001$).

Effect of Route Previewing (Hypothesis 2)

Participants' overall route previewing skills were assessed through self-reports using a 5-point Likert scale (1 = *poor*; 2 = *moderate*; 3 = *good*; 4 = *very good*; 5 = *excellent*) during the interviews conducted prior to data collection. In addition, two further climbing experts, each with climbing qualifications (European Qualifications Framework: $\geq$ Level 5) and 15 years of scientific expertise in the field, were tasked with evaluating the participants' gaze behavior during the preview of the boulder. Consistent with previous research in the field (Medernach, Sanchez, et al., 2024; Mitchell et al., 2020), the experts assessed both the duration of the preview and the number of fixations at climbing holds.

Preview duration refers to the amount of time climbers spent visually processing the boulder during the preview phase. As such, route previewing times reflect how much time climbers invested in

interpreting the climbing movements and developing climbing solutions before actually climbing (Medernach, Sanchez, et al., 2024).

Fixations at climbing holds were assessed to evaluate the total time participants spent fixating at climbing holds and to determine climbing holds without fixations. A fixation was defined as the period during which the eye remained steadily focused on the same hold, within 1° movement tolerance and for a duration equal to or greater than 120 ms (Catteeuw et al., 2009; Savelsbergh et al., 2005). As total absolute fixation times are affected by the overall preview duration, relative fixation times at holds were calculated as a percentage of the total previewing time.

The assessment of climbing holds without fixations was conducted to determine whether participants had overlooked any holds during the preview, in particular the hidden Hold 8, which played a critical role in the more creative option. In cases where no fixation was recorded at a given hold, participants were asked during the postclimbing interviews to confirm that they had indeed overlooked the respective hold. In addition to assessing total fixations at holds and identifying holds without fixations, the duration of fixations at each individual hold was examined. This analysis aimed to identify the areas of interest attended to by participants during route previewing.

Gaze behavior was assessed using a portable, head mounted eye-tracker (ASL Mobile Eye-XG). The device comprised two high-resolution digital cameras (sensor resolution: $1,600 \times 1,200$; horizontal visual range: 60° ; vertical visual range: 40°), mounted on lightweight eyeglasses. The first camera recorded scenery images at a sampling rate of 30 Hz, while the second, monocular eye camera tracked eye movements by detecting the pupil position and corneal reflex through an infrared reflective mirror. The point of gaze was displayed as a cursor on the image of the scene camera. Gaze position was recorded as a sequence of frames at a speed of 30 Hz and with an accuracy of 0.5° . A frame was considered valid if the gaze position was captured within the scene image. The eye-tracking ratio, indicating the percentage of success in tracking the point of gaze, was $94.7 \pm 2\%$. Data were transmitted to a portable wireless data transmit unit and analyzed frame-by-frame using ASL EyeVision software.

A nine-point calibration was conducted before each preview to ensure the accuracy of the eye-movement recording. Participants were instructed to fixate at nine visual dots positioned across the bouldering wall, with calibration points located at the outermost edges of their visual field. During the preview of the boulder, participants were required to remain within a designated area 2 m from the wall. This setup provided realistic viewing conditions (Hüttermann et al., 2018; Roca et al., 2014), allowing an optimal viewing angle of all climbing holds, while also ensuring participants remained within the eye-tracker's recordable visual field.

Postclimbing Interviews

In the initial question, participants were asked (coded as 0 = no; 1 = yes) whether they had a precise climbing solution prior attempting the boulder (*Did you have a precise solution for climbing the boulder?*). In the second question, they had to indicate on a 5-point Likert scale (1 = *very easy*; 5 = *very difficult*) how difficult they found it to develop a climbing solution during the preview (*How difficult was it for you to develop a climbing solution?*). In the third question, they were asked to confirm (coded as 0 = no; 1 = yes) whether they had overlooked those climbing holds for which no fixations were detected by the eye-tracking software (*Did you overlook this hold during the preview?*). In the final question,

they were asked (coded as 0=no; 1=yes) whether they had identified the more creative option (*Did you notice that there was an alternative climbing option that involved shortcutting the boulder by balancing from the starting position to the right?*).

Statistical Analyses

Statistical analyses were performed using IBM SPSS Statistics 29. A priori power analysis was performed using G*Power. Data are presented as percentages or as mean values (M) with SD . An alpha level of $p < .05$ (two-tailed) was used to determine statistical significance. To minimize the impact of multiple testing, multivariate analyses of variance were conducted to determine group differences in the following overarching categories of variables: (a) personal characteristics (age, height, weight); (b) climbing expertise (highest and current IRCRA scores, years of climbing, number of competitions); (c) climbing performance and time management (top, highest hold reached, number of attempts, average climbing time, average rest time); (d) creativity (creativity score, use of the creative option, attempts before using the creative option); and (e) route previewing skills and behaviors (previewing skill, preview duration; absolute and relative fixations, holds without fixations, relative fixations per hold).

An a priori power analysis for the multivariate analyses of variance indicated that a total sample size of $n=28$ was required, assuming a medium to large effect size of $f=0.31$, a statistical power of .8 ($1-\beta$), and an alpha level of .05. The justification for the sample and effect sizes was based on recent research in Olympic bouldering, which used similar data collection procedures and reported substantial effect sizes with comparable sample sizes ($\eta^2 \geq .27$; see Medernach, Henz, et al., 2024). Eta squared is reported as a measure of effect size, with $\eta^2 = .01$ (small effect), $\eta^2 = .06$ (medium effect), and $\eta^2 = .14$ (large effect). Levene's test was conducted to assess the assumption of homogeneity of variance, and Welch's test was applied in cases where this assumption was violated.

In addition to the multivariate analyses of variance, independent t tests were conducted to determine between-group differences for each individual variable. The nonparametric Mann-Whitney U test was used when the assumptions for t tests were violated. Cohen's d was calculated and converted to r for indicating the effect sizes between the groups, with $.10 \leq r < .30$ (small effect), $.30 \leq r < .50$ (medium effect), and $r \geq .50$ (large effect). The Pearson correlation coefficient (r) was calculated to determine a significant relation between two variables. The intraclass correlation coefficient and Cohen's κ were used as statistical measures to assess the level of agreement between two or more raters.

Results

Personal Characteristics

No significant differences in personal characteristics were found between the two study groups, with Wilk's Lambda $\Lambda = 0.975$, $F(3, 24) = 0.204$, $p = .893$, $\eta^2 = .025$ (see Table 1). Specifically, participants in the ELI and ADV groups were of similar age and had comparable body heights and weights.

Climbing Expertise

Between-group comparisons revealed significant differences in the climbing expertise of the participants, with Wilk's Lambda $\Lambda = 0.069$, $F(4, 23) = 77.020$, $p < .001$, $\eta^2 = .931$ (see Table 1).

In particular, the ELI group had higher IRCRA scores (highest and current climbing levels), more years of climbing experience, and had participated in more climbing competitions compared to the ADV group.

Climbing Performance and Time Management

Significant differences were also observed in the climbing performances of the two groups, with Wilk's Lambda $\Lambda = 0.180$, $F(5, 15) = 13.711$, $p < .001$, $\eta^2 = .820$ (see Table 2). The ELI group was more successful in completing the boulder, reached higher holds in their best attempts, and made fewer attempts compared to the ADV group. In contrast, the two groups showed no differences in their time management while attempting the boulder, with similar average climbing times and comparable average rest periods between attempts.

Effect of Climbing Level (Hypothesis 1)

Results revealed significant differences in the creativity variables examined across the groups, with Wilk's Lambda $\Lambda = 0.691$, $F(2, 19) = 4.258$, $p = .030$, $\eta^2 = .309$ (see Figure 2). Specifically, the ELI group demonstrated higher creativity scores than the ADV group. In this context, results indicated a positive correlation between the participants' creativity scores and their IRCRA scores ($r = .654$, $p < .001$). In addition, while all participants in the ELI group (100%) used the creative option to climb the boulder, only about half of the ADV (57%) did so. In this context, results indicated a positive correlation between the participants' use of the creative option and their ability to successfully complete the boulder ($r = .649$, $p < .001$). Specifically, 77.3% of those who used the creative option successfully completed the boulder, whereas none of those who chose the more evident option were able to do so. Furthermore, the ELI group also made fewer attempts before adjusting their climbing strategy and using the creative option, though between-group differences were not statistically significant.

Effect of Route Previewing (Hypothesis 2)

Results revealed significant differences between the groups in their route previewing skills and behaviors, with Wilk's Lambda $\Lambda = 0.226$, $F(6, 21) = 11.997$, $p < .001$, $\eta^2 = .774$ (see Table 3). Specifically, the ELI group reported higher self-perceived previewing skills and had shorter preview durations than the ADV group. In addition, while the ELI group had shorter absolute fixation durations at holds (measured in seconds), both groups did not differ in their total relative fixations (total duration of fixations relative to preview time). However, the ELI group had fewer holds without fixations and, in particular, overlooked the hidden Hold 8 less often than the ADV group.

With regard to the specific areas of interest of the participants, relative fixations at individual holds differed significantly between the groups, with Wilk's Lambda $\Lambda = 0.292$, $F(12, 15) = 3.034$, $p = .023$, $\eta^2 = .708$ (see Figure 3). In particular, elite climbers had higher relative fixations at key holds associated with the creative option (Holds 7 and 8) and the final section of the boulder (Holds 9–12), while advanced climbers demonstrated higher relative fixations at key holds of the more apparent option (Holds 2 and 4–6).

Postclimbing Interviews

Participants in the ELI group more often reported having developed a precise solution for climbing the boulder during the preview (0.9 ± 0.4) than those in the ADV group (0.4 ± 0.5 , $U = 56.0$,

Table 2 Climbing Performances and Time Management of ADV and ELI Groups

Variable (unit)	ADV group (n = 14)	ELI group (n = 14)	t/z	p	r
Tops (number)	0.3 ± 0.5	0.9 ± 0.3	$z = -3.420$	<.001	.641
Highest reached hold (number)	7.2 ± 3.7	11.9 ± 0.5	$z = -3.529$	<.001	.665
Attempts (number)	4.9 ± 3.1	2.4 ± 1.2	$z = -2.190$	.029	.470
Average climbing times (s)	44.5 ± 55.7	49.2 ± 18.0	$t = -.304$	.765	.057
Average rest times (s)	20.1 ± 8.7	24.2 ± 11.9	$t = -.890$	.386	.193

Note. Results are reported as mean ± SD, along with the *t*-test (*t*) or Mann–Whitney *U*-test (*z*) results, the *p* values, and the effect sizes (*r*). ADV = advanced; ELI = elite.

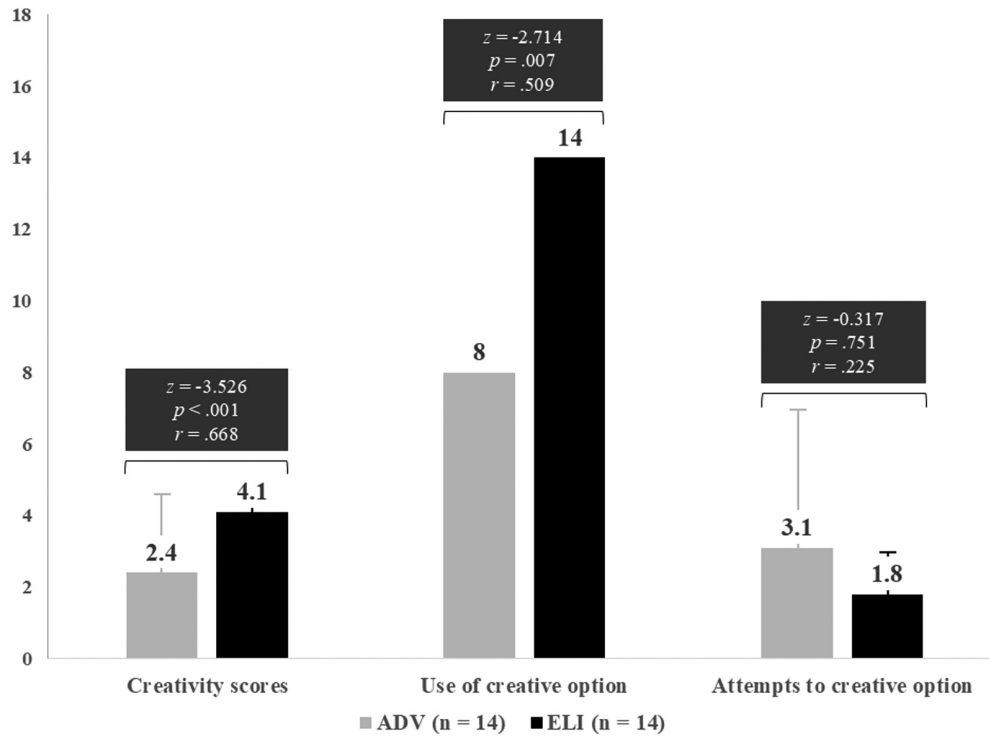


Figure 2 — Creativity of the advanced (ADV) and the elite (ELI) group. Note. The figure illustrates the creativity scores for both groups (1 = marginally creative; 5 = highly creative), along with the number of participants who used the creative option and the mean values for the attempts in which they switched to this option. The boxes above the bars display the Mann–Whitney *U* test (*z*) results, the *p* values, and the effect sizes (*r*).

$z = -2.324$, $p = .020$, $r = .482$). In addition, the ELI group found it less challenging to develop a climbing solution (3.0 ± 0.8) compared to the ADV group (4.1 ± 0.9 , $U = 41.0$, $z = -2.742$, $p = .006$, $r = .543$). Furthermore, results revealed absolute consistency between the climbing holds that participants confirmed to have overlooked and those for which no fixations were detected by the eye-tracking software ($\kappa = 1.000$, $p < .001$). Lastly, advanced climbers more often reported having failed to identify the creative option (0.4 ± 0.5) compared to elite climbers (0 ± 0 , $U = 56.0$, $z = -2.714$, $p = .007$, $r = .512$).

Discussion

The present study explored the role of climbing level and route previewing as contributing factors to climbers' creativity and their ability to find creative climbing solutions. Advanced and elite climbers were tasked with climbing an Olympic boulder that offered two climbing options: a visually apparent and easily

identifiable option that exceeded their climbing levels, and a creative option that was less apparent, yet feasible for both groups and thus more effective. The results revealed that elite climbers were more successful in completing the boulder, attained higher creativity scores, and more often used the creative option. Furthermore, during route previewing, elite climbers had fewer holds without fixations and more fixations at key holds associated with the creative option, despite shorter preview durations and shorter absolute fixation times at holds. Overall, these findings indicate that creativity in Olympic bouldering is associated with climbing level and strategic behavior during route previewing. The subsequent sections discuss the findings in relation to the hypotheses.

Effect of Climbing Level (Hypothesis 1)

A key finding of this study was that elite climbers attained higher creativity scores and more often used the creative option than advanced climbers. These results support Hypothesis 1 and provide the first empirical evidence of a positive relationship between

Table 3 Previewing Skills and Behavior of the ADV and ELI Groups

Variable (unit)	ADV group (n = 14)	ELI group (n = 14)	t/z	p	r
Previewing skills (scores)	2.9 ± 0.9	4.1 ± 0.7	$z = -3.340$	<.001	.597
Previewing duration (s)	111.6 ± 11.9	66.4 ± 25.1	$t = 6.094$	<.001	.755
Absolute fixations at holds (s)	77.0 ± 17.0	44.3 ± 13.4	$t = 4.378$	<.001	.730
Relative fixations at holds (%)	63.0 ± 15.3	68.7 ± 14.6	$z = -0.735$	.462	.187
Holds without fixations (number)	0.9 ± 1.0	0.2 ± 0.6	$z = -2.751$	.006	.391
Hold 8 overlooked (number)	0.6 ± 0.5	0.1 ± 0.4	$t = -2.660$	.008	.483

Note. Results are reported as mean ± SD, along with the *t*-test (*t*) or Mann–Whitney *U*-test (*z*) results, the *p* values, and the effect sizes (*r*). ADV = advanced; ELI = elite.

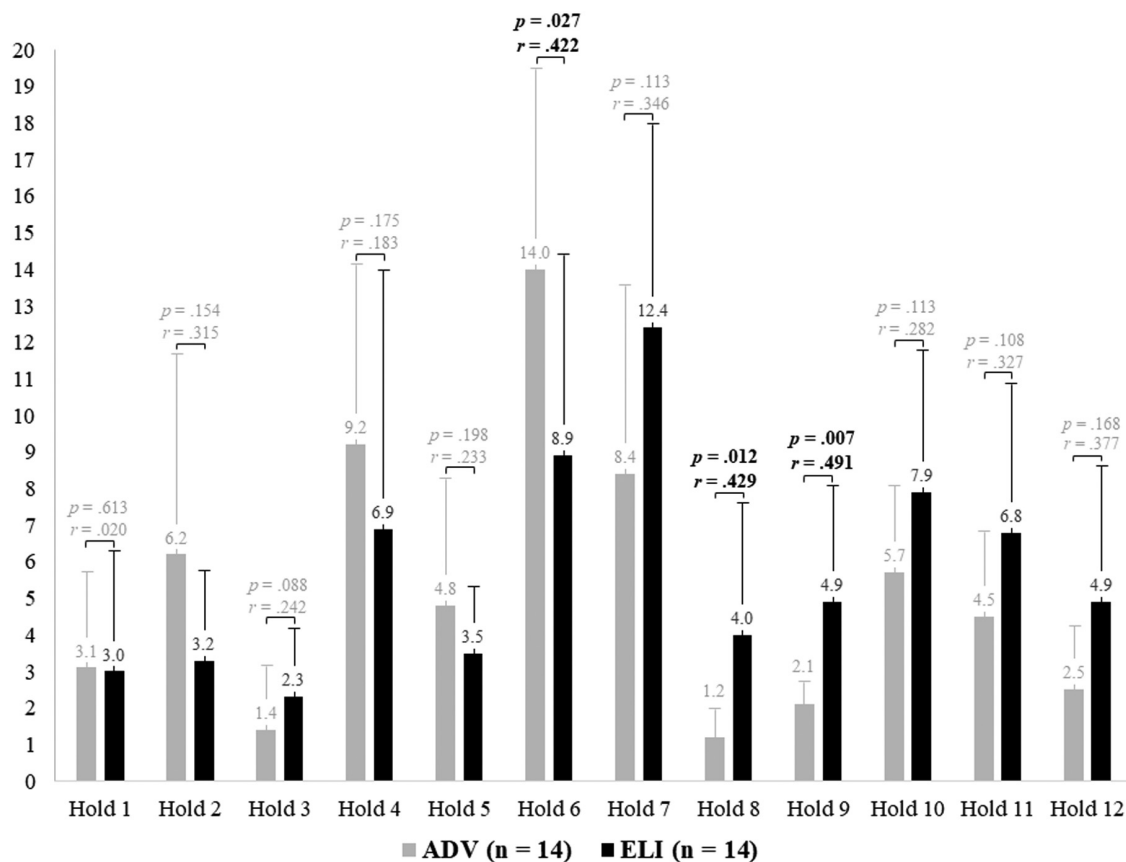


Figure 3 — Hold fixations of the advanced (ADV) and elite (ELI) groups relative to previewing duration. Note. The figure shows the proportion of time participants spent fixating at the different holds, expressed as a percentage of the total previewing duration. The *p* values and effect sizes (*r*) are displayed above the bars.

climbing level and climbers' creativity. Furthermore, the findings align with research in other sports, which highlights a significant relationship between creativity, skill level, and domain-specific expertise (Memmert & Roca, 2019; Roca et al., 2018; Roca et al., 2021). Additionally, the results are consistent with prior research in Olympic bouldering, indicating that successful climbing performance is associated with efficient decision making (Medernach & Memmert, 2021), the generation of appropriate strategic climbing solutions (Medernach, Henz, et al., 2024), and the identification of creative climbing solutions (Künzell et al., 2021).

When considering potential factors that could explain why elite climbers more often relied on the creative option, differences in morphology can be excluded as a potential factor, given that both

groups had comparable height and weight. Moreover, it is noteworthy that about half of the advanced climbers were able to identify and successfully execute the creative solution. This suggests that the creative option was feasible in terms of difficulty and movement complexity for both elite and advanced climbers, thereby corroborating the difficulty level proposed by the expert routesetters. In this context, the positive correlation between climbers' use of the creative option and their success in completing the boulder further substantiates that relying on this option was not only a more original, but also a more efficient solution for climbing the boulder. Consequently, it is unlikely that advanced climbers deliberately avoided the creative option because its movement demands exceeded their technical or motor skills.

Instead, the finding that all advanced climbers who did not use the creative option confirmed that they had failed to identify it suggests that advanced climbers did not intentionally avoid the creative option, but were more likely unable to identify it as a climbing solution. Additional insights from the postclimbing interviews support this assumption, as advanced climbers less often reported having developed a precise climbing solution and found it more challenging to identify a solution for climbing the boulder. Conversely, elite climbers were better at identifying the more creative solution and were consequently less likely to be misled by the more apparent, but less effective option.

A cognitive factor likely underlying the positive relationship between climbing level and climbers' creativity is grounded in the climbing movement repertoire paradigm, which has recently gained research interest in the context of Olympic bouldering (Henz et al., 2025b; Medernach et al., 2025b). Building on Anderson (1983) adaptive control of thought theory, this paradigm posits that expert climbers draw upon an extensive repertoire of movements stored in long-term memory, consisting of declarative knowledge organized within a semantic network; these high-level knowledge structures enable experts to efficiently process visual sensory input, recognize meaningful movement patterns, and anticipate climbing movements by integrating sensory input (e.g., hold configuration) with movement patterns stored in long-term memory (Medernach et al., 2023). Evidently, climbers' movement repertoires also shape their route previewing behavior (Medernach, Sanchez, et al., 2024), which will be further discussed in the subsequent sections.

Effect of Route Previewing (Hypothesis 2)

In addition to climbing level, the findings of this study suggest that climbers' strategic behavior during route previewing plays a critical role in their ability to identify creative climbing solutions. To begin, elite climbers were characterized by superior route previewing skills and allocated less time to previewing the boulder compared to advanced climbers. These findings are consistent with previous research on route previewing, which describes the expertise-processing paradigm as a key mechanism underlying faster information processing during route preview (Medernach, Sanchez, et al., 2024; Medernach & Memmert, 2021). In short, this paradigm builds upon the climbing movement repertoire framework, proposing that experts more rapidly decode sensory information and interpret climbing movements by drawing on their extensive movement repertoire. This, in turn, contributes to shorter previewing and decision making times before physically attempting boulders (Medernach, Henz, et al., 2024).

Recent research on the expertise processing paradigm has revealed that expert climbers engage in shorter, but more concise visual inspections of climbing movements and demonstrate superior recognition of representative movement patterns during the previewing phase (Medernach, Sanchez, et al., 2024). These superior perceptual and cognitive skills observed in experts are grounded in the neural efficiency theory originally proposed by Haier et al. (1988), which posits that experts benefit from more efficient patterns of neural activity when performing cognitive tasks (Del Percio et al., 2009; Immink et al., 2020). In fact, cognitive-motor expertise induces neuroplastic adaptations in neural structures involved in movement planning and execution, resulting in decreased cortical activation during sensory processing and motor task performance (Hatfield, 2018; Tremblay et al., 2021). The neural efficiency theory offers a framework for interpreting Fitts and Posner (1967) model of motor learning and skill acquisition. In

the initial cognitive stage, motor performance requires considerable cognitive effort and attentional resources, often exceeding human processing capacity. Conversely, during the final autonomous stage, movements become automatized, enabling experts to perform tasks more efficiently while using fewer neural resources.

The expertise processing paradigm is also likely to explain that elite climbers had fewer holds without fixations and, in particular, overlooked the hidden Hold 8 less often than advanced climbers. In this context, the absolute consistency between data obtained from the eye-tracking software and verbal reports from the participants confirms that climbers did not intentionally ignore particular climbing holds, but indeed failed to perceive them while previewing the boulder. Given the pivotal role of Hold 8 in the creative option, the fact that advanced climbers more often overlooked this hold is likely a major reason why they were also less able to identify this solution. Findings related to gaze patterns suggest that their failure to perceive Hold 8 and identify the creative option can be attributed to differences in attentional focus during the previewing phase. In fact, elite climbers had higher relative fixations at key holds associated with the creative option, including Hold 8. This finding further supports the assumption that elite climbers were less likely to be misled by the more difficult option, probably due to their superior climbing expertise, which allowed them to more accurately estimate the feasibility of this climbing solution. Instead, they directed more attention toward holds critical for identifying the creative option.

In contrast, advanced climbers spent a higher amount of their preview fixating at the most difficult section of the easily identifiable option. Although focusing on the most difficult section during route previewing is a common and effective strategy among climbers (e.g., Sanchez et al., 2019), this approach likely contributed to a narrower attentional focus in the present study. This may explain why advanced climbers were more prone to missing critical information necessary to identify the creative option, such as Hold 8. Indeed, in line with the inattentional blindness framework (Rock et al., 1992) and consistent with research from other sports (e.g., Memmert & Furley, 2007; Roca et al., 2018), focusing on a specific aspect of the boulder is likely to have limited the amount of information that climbers were able to perceive during the previewing phase, given the limited time available for previewing the boulder.

Overall, the findings of the present study support Hypothesis 2 and suggest that, alongside an extensive movement repertoire, which is essential for effectively processing visual sensory input and identifying potential climbing solutions, climbers' behavior during route previewing constitutes a critical factor contributing to the identification of creative motor solutions. Building on Guilford (1967) framework of divergent thinking, the generation of creative solutions requires cognitive flexibility to explore novel, yet effective approaches (e.g., Fardilha & Allen, 2020; Memmert & Roca, 2019; Runco & Acar, 2019). Consistent with findings within the perceptual-cognitive framework in other sports, the ability to identify meaningful patterns (Roca et al., 2021), allocate attention to task-relevant features (Hüttermann et al., 2014), and effectively filter out irrelevant distractions (Furley et al., 2013), along with maintaining a broad attentional focus during problem-solving (Memmert & Roca, 2019; Roca & Williams, 2016), is essential for the efficient processing of task-related information and the identification of potential creative solutions (Furley et al., 2010; Memmert & Roca, 2019; Moraru et al., 2016). Conversely, converging on a single response—such as mainly focusing on the most challenging section of the boulder—increases the risk of

inattentional blindness and contributes to a narrowed attentional breadth, thereby limiting the perception of relevant environmental information that could support the identification of creative motor solutions (Memmert & Furley, 2007; Roca et al., 2018).

In terms of practical implications, the expertise processing paradigm posits that effective route previewing arises from long-term expertise and deliberate practice (Medernach, Sanchez, et al., 2024), which Ericsson et al. (1993) defined as purposeful, focused, and effortful practice aimed at the systematic enhancement of skills over time. Building upon the movement repertoire paradigm, training regimens in Olympic bouldering should encompass climbing movement versatility to expand the climber's movement repertoire, which is essential for efficiently processing visual sensory input, recognizing meaningful movement patterns, and anticipating climbing movements (Henz et al., 2025b; Medernach et al., 2025b). Beyond deliberate variation in training content and movement demands, climbers should engage in playful exploration of novel climbing movements and be encouraged to make autonomous decisions while pursuing creative problem-solving strategies (Marshall et al., 2024; Memmert & Roca, 2019). Such an environment cultivates unconventional thinking and cognitive flexibility, both of which are particularly relevant during route previewing to broaden climbers' attentional focus.

From a critical perspective on experimental limitations, the present study involved only a single boulder with two climbing options, thereby constraining solution variability and limiting opportunities for creative expression. While this investigation is the first empirical exploration of creativity in Olympic bouldering, further research incorporating a broader range of boulders and including female climbers is necessary to more comprehensively understand the factors underlying creativity in Olympic bouldering. Building on recent findings that elite climbers use fewer scans and more linear scan paths during route previewing compared to less skilled climbers (Medernach, Sanchez, et al., 2024), future research could more thoroughly examine how scan patterns influence climbers' ability to identify creative motor solutions. Furthermore, given the positive association identified between creativity and climbing level, subsequent research might investigate whether climbers of comparable ability exhibit differences in creativity and elucidate the factors contributing to such variability. Such research could also use less structured, more open-ended follow-up interviews to gain deeper insights into climbers' perceptions and decision making processes.

Conclusion

The findings of this study indicate that creativity in Olympic bouldering is positively associated with both climbing level and strategic behavior during route previewing. The relationship between climbing level and climbers' creativity is likely attributable to the climbing movement repertoire paradigm, which posits that expert climbers draw upon an extensive repertoire of movement patterns stored in long-term memory to efficiently process visual sensory input, recognize meaningful movement patterns, and anticipate climbing movements. Alongside an extensive movement repertoire, differences in route previewing behavior can be attributed to the expertise processing paradigm, which likely accounts for more effective attentional focus and enhanced processing of visual information among elite climbers during route previewing. Given that the generation of creative solutions requires cognitive flexibility, maintaining a broad attentional focus during route previewing is essential, as it enables climbers to perceive a

wide range of information about boulders. This reduces the likelihood of inattentional blindness and the risk of overlooking critical information relevant to creative motor solutions, thereby enhancing climbers' ability to identify creative climbing solutions.

Transparency and Openness

To adhere to the Transparency and Openness Promotion guidelines, data are accessible (<https://doi.org/10.7910/DVN/U0BI2K>) for the purpose of reproducing the results or replicating the procedure.

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